

Population Aging in Advanced and Emerging Economies: Capital Flows and Fiscal Spillovers*

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Abstract

Advanced economies have started aging much earlier than emerging economies and did so at a much slower pace. This study investigates the fiscal implications to emerging countries of advanced countries' earlier aging and seeks to understand the fiscal impact of emerging countries' fast catch-up. I estimated the long-run fiscal effects using a Panel FMOLS approach on a panel of advanced and emerging economies, it finds that the latter benefited from the earlier aging in the former starting in the 1990s. It also finds that aging driven more by an increase in longevity will have a lower fiscal impact. Going forward, as the emerging countries age and catch up with the advanced, they will be hit by the double-whammy of rising deficits and the abatement of capital flows, which will critically tighten their fiscal space.

Keywords: Life expectancy, Population growth, Demographic transition, Interest-growth differential, Fiscal space, Debt dynamics

JEL Classifications: E62, F41, H63, J11

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1 Introduction

Across the globe, a pronounced shift in the demographic landscape is unfolding, characterized by a decline in fertility rates and an extension of life expectancy. This worldwide trend was initially distinct between emerging and advanced economies, with advanced nations experiencing these changes much earlier in the postwar era. However, emerging economies are already going through the same changes, though at a much faster pace.

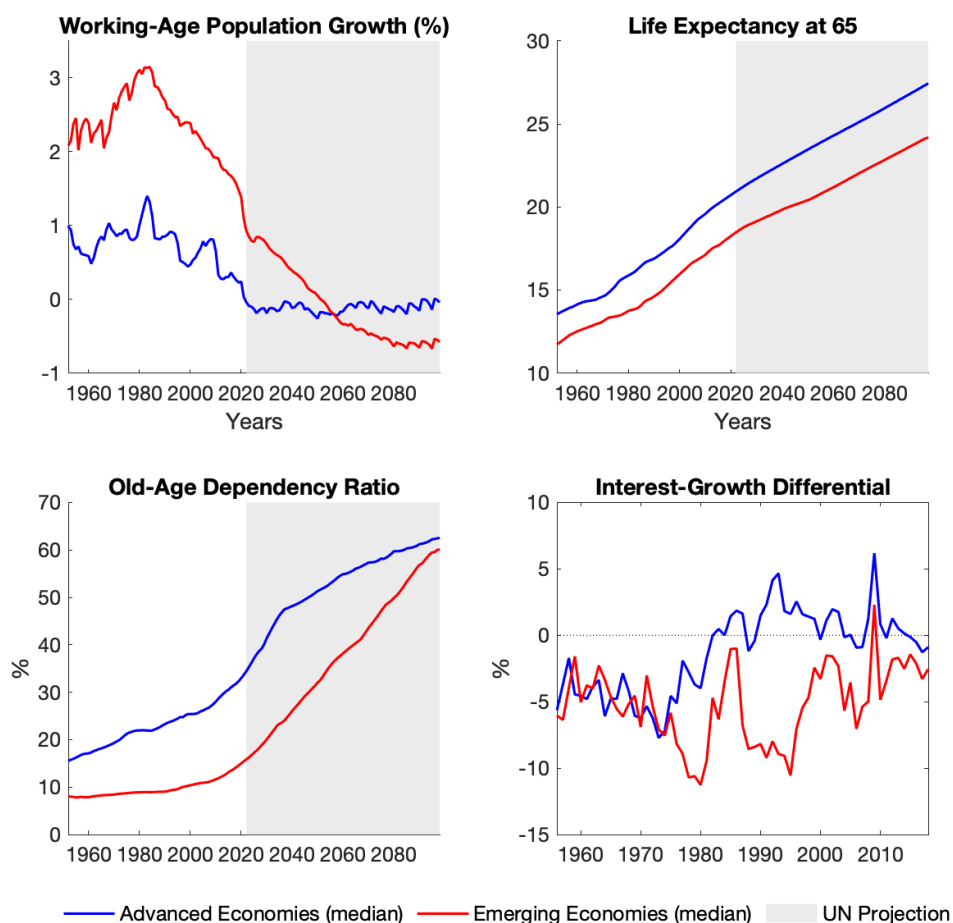
It took seven decades for the old-age population (65 years or older) in a set of advanced economies to increase from 15% of the working-age population to 35% (1950-2021). Conversely, emerging economies are projected to undergo this transformation in just 30 years (2019-2049) ([United Nations, 2022](#)). Given the significant fiscal ramifications of an aging population, the key questions that arise are: what are the fiscal impacts of the fast demographic change emerging economies are going through? Does the desynchronized character of this demographic change in emerging economies have any bearing to their fiscal dynamics?

The most apparent economic consequence of demographic shifts is the heightened pressure on pension systems. As the working-age population contracts, contributions to these systems decline, while an aging population results in increased disbursements. In this regard, the experience of emerging economies will be similar to that of advanced economies in that the burden on their pension systems will be increasingly higher. Unlike advanced economies, however, emerging nations face an accelerated pace of aging, which significantly shortens the time frame for effective reforms in countries that on average have less fiscal credibility.

In addition, changes in factor prices and broader general equilibrium effects are important to evaluate the total fiscal cost of these changes. As [Attanasio et al. \(2007\)](#) has showed, projections should take into account the difference in demographic trends between regions in order to gauge the full fiscal implications of population aging. As advanced nations aged early on, the increased savings rates to finance a longer period in retirement and the reduced rate of return on investments due to a slower growing workforce both created supportive conditions for capital outflows towards the developing younger economies. These economies benefitted from these capital flows as they meant a downward pressure on their cost of capital, in particular on the interest rate on government debt. As these emerging economies age and catch up with the more advanced ones, these supportive conditions will gradually disappear and the cost of government debt will increase as a result, implying a second negative effect these nations will be subject to. An effect that is not taken into account if one abstracts from global developments.

Examining Figure 1 gives us important insights into this backdrop. The top panels in Figure 1 show the evolution of the two main demographic drivers of population aging across

Figure 1: Demographics, Capital Flows and Debt Dynamics



Note: Top left panel: Cross-sectional median of old-age dependency ratio for 23 advanced and 23 emerging economies between 1956 and 2100. Top right panel: Cross-sectional median of working-age population growth in advanced minus cross-sectional median in emerging economies (green); cross-sectional median of life expectancy at 65 in advanced minus cross-sectional median in emerging economies (purple) between 1956 and 2100. Bottom left panel: foreign capital flows to emerging economies are defined as the net change in liabilities to nonresidents in terms of foreign direct investment and portfolio investment. Bottom right panel: Cross-sectional median old-age dependency ratio for 23 advanced and 23 emerging economies between 1956 and 2100. Source: United Nations World Population Prospects, 2022 Revision (top panels); IMF (bottom left); [Mauro and Zhou \(2021\)](#).

time: the working-age population growth rate, a measure of fertility, and life expectancy at 65 years old, a measure of longevity. The graph on the upper left corner, displaying the evolution of the median growth rate of the working-age population, immediately draws attention to the fact that since the post-war era the working-age population has grown at a much faster rate in developing economies, but that is set to end as these economies are now witnessing a collapse

in fertility and are seeing their population growth rates plummet.

The upper right graph shows the increase in median life expectancy at 65 years in both advanced and emerging economies. Even though emerging economies present a shorter life expectancy at old-age as compared to their counterparts, the increase in life expectancy in both regions happens almost in parallel, likely due to advances in technology that are more easily exported across countries.

The lower left panel shows how the combination of lower fertility and higher longevity has affected advanced economies and have recently started to affect the emerging economies too. Advanced economies have seen a gradual rise in this metric, presumably leading to a higher fiscal burden as social support systems like pensions and healthcare get strained. This earlier demographic shift also influenced advanced economies to become net exporters of capital as their domestic investment opportunities diminished and households increased savings to finance a longer retirement. Emerging economies, meanwhile, are seeing this ratio escalate in a much shorter period, resulting in a sudden, sharp increase in fiscal burdens, thus compressing the time available for policy adjustments.

Finally, the lower right graph presents the median interest-growth differential between the two groups. Up until the mid-1970s, the differential remained fairly similar for both groups. However, the gap widened as advanced economies started experiencing higher differentials while their emerging counterparts saw a decline. This could be attributed to earlier demographic changes in advanced economies that funneled capital toward emerging markets. But as these markets now undergo rapid aging, we can expect the interest-growth differentials to converge, posing challenges for emerging economies. This will likely impact their appeal to foreign capital, pushing up borrowing costs and further complicating fiscal adjustments.

This paper seeks to investigate whether the disparity in demographic trends between advanced and emerging economies has led to a loosening of fiscal conditions in emerging economies and what the eventual convergence in these trends means for these conditions. In doing so it also makes two secondary contributions. First, it shows that changes in the ratio between benefits and contributions to the pension system is not a sufficient statistic to evaluate the impact of population aging. Instead, the source of aging matters, whether it is an increase in longevity or a fall in fertility. Second, it establishes analytically and empirically how demographic change affect the interest-growth differential, the rate at which the government debt-to-GDP ratio grows when the primary balance is zero.

Literature Review: This paper is related to a few strands of the literature. The first is the one that applies large scale OLG models to study the effects of the demographic transition on a set of economic variables in the tradition of the seminal paper or [Auerbach et al. \(1987\)](#). The topic of this study is close to [Attanasio et al. \(2007\)](#)'s. The authors argue that social

security reform needs to consider the difference in demographic trends across countries but they maintain their focus on the case of advanced economies. [Krueger and Ludwig \(2007\)](#) use an open economy model to quantify the impact of the demographic transition in factor prices, capital flows and welfare; [Vogel et al. \(2017\)](#) also use a multi-country OLG model to address policy responses to the aging process, and [Gagnon et al. \(2021\)](#) use the same methodology to account for the decline in output growth and real interest rates in the United States. These papers feed the demographic transition as an exogenous process to the model to explain and project the path of macroeconomic aggregates under different policy scenarios. The results of simulations, however, are particular to the set of parameters and demographic change fed into the model. This paper, in contrast, looks for a general result and also seeks to identify the role of each demographic driver.

A second strand of the literature is the one that uses open-economy OLG models to emphasize the importance of the different demographic dynamics across countries. In this regard, this study is very close to [Bárány et al. \(2023\)](#), who also explores the effects of demographic change in shaping capital flows between advanced and emerging economies. Other previous studies on the topic are [Bryant \(2004\)](#), who provides a broad but non-technical overview, [Feroi \(2003\)](#), who find that demographic differences help explain much of the observed capital flows among the G-7 economies, and [Ferrero \(2010\)](#), who employs a tractable open-economy OLG model to decompose the US trade balance and find an important role for demographics.

Finally, a third branch of the literature is the one that studies the interest-growth differential. [Blanchard and Weil \(2001\)](#), studies theoretical implications of dynamic (in)efficiency to debt Ponzi games under uncertainty, while more recently [Blanchard \(2019\)](#) studies the fiscal and welfare costs of public debt in an environment of negative interest-growth differentials. [Reis \(2021\)](#) in turn seek to establish a limit to the primary deficit in a context of negative interest-growth differential on public debt. However, these papers only analyze the consequences of the sign of the interest-growth differential for fiscal policy. They do not explain what drives the changes in the differential across time.

The present paper thus seeks to fill the aforementioned gaps by investigating the role demographic changes have in impacting debt dynamics through the interest-growth differential.

I resort to a tractable 2-country Overlapping Generations model a la [Diamond \(1965\)](#) stripped down to the most fundamental macroeconomic relationships to investigate the main channels through which the demographic processes affect fiscal variables.

The analytical results show that earlier aging in advanced countries led to increased private savings and reduced investment returns, which are supportive of capital flows to young emerging countries. The influx of capital depresses interest rates and stimulates economic

growth, making public debt rollover less costly. These are called fiscal spillovers and they exist as long as the demographic gap exists. As the young emerging country ages the gap closes, capital inflows decline and fiscal conditions tighten.

The rest of the paper is structured as follows: Section 2 presents a two-country OLG model, Section 3 derives the theoretical results and discusses how they help interpret the effects of aging in emerging economies, and Section 4 concludes the paper.

2 Simple Model

The following model is a two-country version of the standard [Diamond \(1965\)](#) OLG model augmented with a Pay-As-You-Go (PAYGO) social security system and a survival probability.

2.1 Households

Households potentially live for two periods. They live as workers during their first period and a share $1 - \delta_{t+1}$ of them dies at the end of that period and do not reach the second period. The exogenous variable δ_{t+1} can be thus considered to be the share of the time period when agents are alive when old. The number of workers in period t is given by N_t and the number of retirees is thus $\delta_t N_{t-1}$. The number of young workers grows at the gross rate g_{Nt} : $N_t = g_{Nt} N_{t-1}$.

Households have log-utility over consumption when young, c_{1t} , and old, c_{2t+1} , and therefore their utility over lifetime consumption is given by the function below:

$$U(c_{1t}, c_{2t+1}) = \ln c_{1t} + \beta \delta_{t+1} \ln c_{2t+1} \quad (1)$$

where β represents the discount factor.

Labor supply is inelastic. Young households receive a wage w_t in exchange for the labor supplied and also pay a lump-sum tax, τ_t . They then consume, c_{1t} , and save, s_t , out of their after-tax labor income. When old, households receive interest over their savings given by the gross interest rate, R_{t+1} , and retirement benefits, e_t .

$$c_{1t} + s_t = w_t - \tau_t \quad (2)$$

$$c_{2t+1} = \frac{R_{t+1}}{\delta_{t+1}} s_t + e_t \quad (3)$$

The first order conditions of this problem imply the following savings function:

$$s_t = \frac{\beta\delta_{t+1}}{1 + \beta\delta_{t+1}}(w_t - \tilde{\tau}_t) - \frac{\delta_{t+1}}{(1 + \beta\delta_{t+1}) R_{t+1}} e_{t+1} \quad (4)$$

2.2 Firms

Firms are assumed to combine labor, L_t , and capital, K_t , to produce the single good in the economy, Y_t , according to the following Cobb-Douglas utility function:

$$Y_t = (A_t L_t)^{1-\alpha} K_t^\alpha \quad (5)$$

with $\alpha \in (0, 1)$ and total factor productivity, A_t , that grows at rate g_{Ait} : $A_t = g_{Ait} A_{t-1}$. As a result, wage and the gross interest rate are paid their marginal products:

$$\begin{aligned} R_t &= \alpha k_t^{\alpha-1} \\ w_t &= (1 - \alpha) A_t k_t^\alpha \end{aligned} \quad (6) \quad (7)$$

where $K_t \equiv k_t / (A_t N_t)$ is the capital stock measured in efficient labor units.

2.3 Government

Assume government's only tax revenue comes from the lump-sum tax levied on workers, while its only transfers comprise the social security benefits paid to retirees. For tractability, retirement benefits, E_t , are a fraction ν of the current period's wage:

$$e_t = \nu w_t \quad (8)$$

The social security budget is not forced to balance every period; the government can issue debt, b_t , if necessary. Therefore the law of motion of public debt is given by:

$$B_{t+1} = E_t + G_t - T_t + R_t B_t \quad (9)$$

where B_{t+1} is the government debt measured at the end of period t and start of period $t + 1$, E_t is the total social security benefits paid in period t , T_t is the total tax revenue collected in period t and R_t is the gross interest rate paid by public debt (which in this model is equal

to the marginal return on capital). Define B_{t+1} as the debt-to-output ratio measured at the end of period t (and start of period $t + 1$): $B_{t+1} \equiv B_{t+1}/Y_t$ and rewrite equation (9) as:

$$b_{t+1} = g_t - \tau_t + \frac{\delta_t \nu (1 - \alpha)}{g_{Nit}} + \frac{\alpha}{g_{Ait} g_{Nit}} \frac{k_{t-1}^\alpha}{k_t} b_t \quad (10)$$

where τ_t is the tax burden $\tau_t \equiv N_t \tilde{\tau}_t / Y_t$.

2.4 Market clearing

The capital market clearing condition is that total savings equal the total capital and debt stocks at the beginning of the following period:

$$N_t s_t + N_t^* s_t^* = K_{t+1} + K_{t+1}^* + B_{t+1} + B_{t+1}^* \quad (11)$$

where variables with the superscript represent the foreign economy. Dividing (11) through by $A_t N_t$ and rearranging yields:

$$\frac{s_t}{A_t} + \varphi_t^* \frac{s_t^*}{A_t^*} = g_{At+1} g_{Nt+1} k_{t+1} + g_{At+1}^* g_{Nt+1}^* \varphi_t^* k_{t+1}^* + b_{t+1} k_t^\alpha + \varphi_t^* b_{t+1}^* (k_t^*)^\alpha \quad (12)$$

where $\varphi_t^* \equiv A_t N_t / (A_t^* N_t^*)$ is the relative size of the foreign economy's effective labor.

Substitute for w_t and e_{t+1} in the savings function (4) and get as a result:

$$s_t = \frac{\beta \delta_{t+1}}{1 + \beta \delta_{t+1}} ((1 - \alpha) A_t (k_t)^\alpha - \tau_t A_t y_t) - \frac{1 - \alpha}{\alpha} \frac{\delta_{t+1}}{1 + \beta \delta_{t+1}} \nu A_t k_{t+1} \quad (13)$$

Plugging equation (13) into (12) yields:

$$\begin{aligned} & \frac{\beta \delta_{t+1}}{1 + \beta \delta_{t+1}} ((1 - \alpha) - \tau_t) (k_t)^\alpha - \frac{1 - \alpha}{\alpha} \frac{\delta_{t+1}}{1 + \beta \delta_{t+1}} \nu k_{t+1} + \\ & \varphi_t^* \left(\frac{\beta^* \delta_{t+1}^*}{1 + \beta^* \delta_{t+1}^*} ((1 - \alpha) - \tau_t^*) (k_t^*)^\alpha - \frac{1 - \alpha}{\alpha} \frac{\delta_{t+1}^*}{1 + \beta^* \delta_{t+1}^*} \nu^* k_{t+1}^* \right) \\ & = g_{At+1} g_{Nt+1} k_{t+1} + g_{At+1}^* g_{Nt+1}^* \varphi_t^* k_{t+1}^* + b_{t+1} k_t^\alpha + \varphi_t^* b_{t+1}^* (k_t^*)^\alpha \quad (14) \end{aligned}$$

Given perfect capital mobility, the no-arbitrage condition holds and $k_t = k_t^* = k_t$. Equation (14) becomes:

$$\begin{aligned}
& \left(\frac{\beta\delta_{t+1}}{1+\beta\delta_{t+1}} ((1-\alpha) - \tau_t) + \varphi_t^* \frac{\beta^*\delta_{t+1}^*}{1+\beta^*\delta_{t+1}^*} ((1-\alpha) - \tau_t^*) \right) k_t^\alpha \\
& - \frac{1-\alpha}{\alpha} \left(\frac{\delta_{t+1}}{1+\beta\delta_{t+1}} \nu + \varphi_t^* \frac{\delta_{t+1}^*}{1+\beta^*\delta_{t+1}^*} \nu^* \right) k_{t+1} \\
& = (g_{At+1}g_{Nt+1} + g_{At+1}^*g_{Nt+1}^*\varphi_t^*)k_{t+1} + (b_{t+1} + \varphi_t^*b_{t+1}^*)k_t^\alpha \quad (15)
\end{aligned}$$

Note that given initial values for the capital stock in efficiency units k_{-1} and k_0 and given the path for the exogenous variables, the path for efficient capital is given by equations (15) and (10).

2.5 Analysis

In this section I carry out the analysis of the effects of demographic shocks on economic variables. In particular, the interest lies on the effect of such shocks on fiscal variables.

I will first analyze the effect of demographic changes on the interest-growth differential and then show that a fall in population growth has the opposite effect of an increase in longevity. The final result will hinge on this distinction.

2.5.1 Interest-Growth Differential

Proposition 1. *A fall in the growth rate of the labor force unambiguously increases the adjusted interest-growth differential next period:*

$$\frac{\partial \frac{R_{t+1}}{g_{Yt+1}}}{\partial g_{Nt+1}} < 0$$

An increase in the survival probability affects the interest-growth differential as follows:

$$\frac{\partial \frac{R_{t+1}}{g_{Yt+1}}}{\partial \delta_{it+1}} \begin{cases} \leq 0 & \text{if } s_t \geq 0 \\ > 0 & \text{if } s_t < 0 \end{cases}$$

Proof. Appendix. □

The results in Proposition 1 hinge on the comparison between the elasticities of the real interest rate and of the output growth rate with respect to the demographic variables. Both elasticities will in turn be functions of the elasticity of capital in efficiency units to demographics.

The elasticity of the real interest rate to the population growth rate will be lower than the elasticity of the output growth rate with respect to that same variable if the elasticity of capital is lower than one (in absolute value). That will be the case if the foreign country is sizeable and/or if there is a PAYGO social security system in place in any country. The existence of a sizeable foreign country will diminish the effect of the domestic country on the total stock of capital, whereas a PAYGO social security system will lead agents to save more in case there is a decrease in the stock of capital in efficiency units, counterbalancing changes in that stock. Both dynamics guarantee an inelastic capital stock in efficiency units with respect to the population growth rate, which leads to the first result: a fall in the population growth rate widens the interest-growth differential.

On the other hand, the increased longevity will work through the savings channel only. Savings will go up if they are positive, as the increased longevity will make agents value more the second period. Now, if agents are borrowing, an increase in longevity will diminish the interest rate adjusted by the survival probability, making consuming more in the first period more appealing. Focusing on the positive savings case, an increase in longevity will increase the stock of capital and lower the real interest rate. By the same token, output growth goes up due to the larger capital stock. The result is a smaller interest-growth differential. If savings are negative, we have the exact opposite scenario as consequence.

Proposition 2. *A fall in the growth rate of the labor force unambiguously increases the adjusted interest-growth differential beyond the following period:*

$$\frac{\partial \frac{R_{t+k}}{g_{Yt+k}}}{\partial g_{Nt+1}} < 0, \text{ for } k > 1$$

An increase in the survival probability affects the interest-growth differential beyond the following period as follows:

$$\frac{\partial \frac{R_{t+k}}{g_{Yt+k}}}{\partial \delta_{it+1}} \begin{cases} \leq 0 & \text{if } s_t \geq 0 \\ > 0 & \text{if } s_t < 0 \end{cases}$$

Proof. Appendix. □

Proposition 3. *The aging of population abroad either as a result of a fall in the growth rate of the labor force or an increase in life expectancy unambiguously decreases the interest-growth differential in the domestic economy in the current period:*

$$\frac{\partial \frac{R_{t+1}}{g_{Yt+1}^j}}{\partial g_{Nt+1}^j} < 0, \text{ for } k > 1$$

Proof. Appendix. □

2.5.2 Fiscal Space

In order to evaluate the fiscal impact of the demographic transition, it is necessary to introduce the concept of fiscal space. To that end it is useful to divide the components of the primary balance into *demographic* components and *remainder* components. The demographic components represent the parts of the primary balance that are a direct function of demographics, whereas the remainder components are simply the primary balance minus its demographic components. The remainder then becomes a function of demographics only *through* the demographic components. The breakdown is as follows:

$$\tau_t = \tau_t^d(g_{Nt}, \delta_t, \mathbf{x}_t) + \tau_t^r \quad (16)$$

$$g_t = g_t^d(g_{Nt}, \delta_t, \mathbf{x}_t) + g_t^r \quad (17)$$

$$e_t = e_t^d(g_{Nt}, \delta_t, \mathbf{x}_t) + e_t^r \quad (18)$$

Now solve equation (10) forward to get:

$$b_t = \sum_{i=0}^{\infty} \left[\left(\prod_{j=0}^i \frac{g_{Y,t+j}}{R_{t+j}} \right) [\tau_{t+i} - g_{t+i} - e_{t+i}] \right] + \lim_{s \rightarrow \infty} \left(\prod_{j=0}^s \frac{g_{Y,t+j}}{R_{t+j}} \right) [\tau_{t+s} - g_{t+s} - e_{t+s}] \quad (19)$$

Assume that the No-Ponzi condition holds and break each component of the primary surplus into demographic and remainder components.

$$b_t = \sum_{i=0}^{\infty} \left[\left(\prod_{j=0}^i \frac{g_{Y,t+j}}{R_{t+j}} \right) \underbrace{[\tau_{t+i}^d - g_{t+i}^d - e_{t+i}^d]}_{\equiv \sigma_t^d} \right] + \sum_{i=0}^{\infty} \left[\left(\prod_{j=0}^i \frac{g_{Y,t+j}}{R_{t+j}} \right) \underbrace{[\tau_{t+i}^r - g_{t+i}^r - e_{t+i}^r]}_{\equiv \sigma_t^r} \right] \quad (20)$$

Consider now a marginal change in an exogenous demographic variable $\varkappa_t = \{g_{Yt}, \delta_t\}$. This change will have a direct impact on the demographic primary surplus, σ_t^d , and will also have an impact on the interest-growth differential, R_t/g_{Yt} . Note that these impacts are also a function of the fiscal policy in place. The decision of how to finance the changes in the demographic primary surpluses will also affect the response of endogenous variables to the exogenous demographic variables. The financing decision will be reflected in the changes in the remainder primary surpluses. These changes will underlie the definition of fiscal space.

Definition 1. A change in fiscal space due to a change in a demographic variable $\varkappa_t$ is defined by the present value of changes in the remainder primary surpluses, s_t^r . If this change is positive, that implies an effort to raise the primary surplus after the demographic change, i.e. there is a loss of fiscal space. Likewise, if the change is negative, then there is creation of fiscal space.

$$\sum_{i=0}^{\infty} \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right) \frac{\partial \sigma_{t+i}^r}{\partial \varkappa_t} \begin{cases} > 0, \text{ loss of fiscal space} \\ < 0, \text{ creation of fiscal space} \end{cases} \quad (21)$$

Now take the first derivative of the current debt-to-GDP ratio (pre-determined) with respect to the demographic variable $\varkappa_t$:

$$\begin{aligned} \frac{\partial b_t}{\partial \varkappa_t} = & \sum_{i=0}^{\infty} \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right) \frac{\partial \sigma_{t+i}^d}{\partial \varkappa_t} + \sum_{i=0}^{\infty} \frac{\partial \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right)}{\partial \varkappa_t} \sigma_{t+i}^d \\ & + \sum_{i=0}^{\infty} \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right) \frac{\partial \sigma_{t+i}^r}{\partial \varkappa_t} + \sum_{i=0}^{\infty} \frac{\partial \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right)}{\partial \varkappa_t} \sigma_{t+i}^r = 0 \end{aligned} \quad (22)$$

The equation above implies that the change in fiscal space can be decomposed in the following two parts. Note that the impact on the interest-growth differential affects the entire primary surplus, not just the demographic part.

$$\begin{aligned} - \sum_{i=0}^{\infty} \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right) \frac{\partial \sigma_{t+i}^r}{\partial \varkappa_t} = & \underbrace{\sum_{i=0}^{\infty} \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right) \frac{\partial \sigma_{t+i}^d}{\partial \varkappa_t}}_{\text{effect on demographic primary surplus}} + \underbrace{\sum_{i=0}^{\infty} \frac{\partial \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right)}{\partial \varkappa_t} \sigma_{t+i}}_{\text{effect on interest-growth differential}} \end{aligned} \quad (23)$$

The decomposition will be key in differentiating the fiscal impact of both components of the demographic transition. The first component is negative given the aging of the population, regardless its source. The second component, however, will be negative when there is a fall in population growth, but positive when there is an increase in longevity, an immediate result of Propositions 1 and 2.

In order to rigorously show the difference, we need to translate the definitions above to their correspondent variables in the model.

$$\tau_t^d = 0 \quad \tau_t^r = \tau_t \quad (24)$$

$$g_t^d = 0 \quad g_t^r = g_t \quad (25)$$

$$e_t^d(g_{Nt}, \delta_t, \mathbf{x}_t) = e_t = \frac{(1 - \alpha)\delta_t \nu}{g_{Nt}} \quad e_t^r = 0 \quad (26)$$

Proposition 4. *A fall in the growth rate of the labor force causes a loss of fiscal space:*

$$\sum_{i=0}^{\infty} \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right) \frac{\partial \sigma_{t+i}^r}{\partial g_{Nt}} > 0$$

If the domestic country is “sizeable enough”, there exists a survival probability $\bar{\delta}_{t+1}$ such that:

$$\sum_{i=0}^{\infty} \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right) \frac{\partial \sigma_{t+i}^r}{\partial \delta_t} \begin{cases} \leq 0 & \text{if } \delta_t \leq \bar{\delta}_t \\ > 0 & \text{if } \delta_t > \bar{\delta}_t \end{cases}$$

Proof. Appendix. □

Proposition 5. *The aging of population abroad either as a result of a fall in the growth rate of the labor force or an increase in life expectancy unambiguously creates fiscal space in the domestic economy in the current period:*

$$\sum_{i=0}^{\infty} \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right) \frac{\partial \sigma_{t+i}^r}{\partial g_{Nt}^*} < 0 \quad \text{and} \quad \sum_{i=0}^{\infty} \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right) \frac{\partial \sigma_{t+i}^r}{\partial \delta_t^*} < 0$$

Proof. This is straightforward and comes as an implication of Propositions 2 and 3. A fall in the growth rate of the working-age population or an increase in life expectancy lead to reductions in the interest-growth differentials and thus:

$$\sum_{i=0}^{\infty} \frac{\partial \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right)}{\partial x_t} \sigma_{t+i} > 0 \quad \implies \quad \sum_{i=0}^{\infty} \left(\prod_{j=0}^{\infty} \frac{g_{Y,t+j}}{R_{t+j}} \right) \frac{\partial \sigma_{t+i}^r}{\partial x_t} < 0 \quad (27)$$

where $x \in [g_{Nt}^*, \delta_t^*]$ □

3 Empirical Analysis

3.1 Data

The data set is a panel of 23 advanced economies¹ and 23 emerging economies² spanning the period 1956-2018. It comes from different sources. The World Population Prospects ([United Nations, 2022](#)) provides the demographic data on working-age and old-age population and life expectancy. To map the data to the model, working age population is defined as 20 to 64 years old. This approximates the retirement age at each country to be constant at 65 years of age, which is not a big issue given that the retirement age does not vary much across time and that what matters ultimately is the growth rate of the working-age population and life expectancy at old age. This is also in line with the literature. Therefore old age is defined as 65 years of age or older. Life expectancy is considered as the remaining life expectancy at 65 years old.

Fiscal variables such as net and gross government debt, primary balance and interest spending are taken from the [Mauro and Zhou \(2021\)](#) database. Data on old-age cash benefits comes from the OECD, and data on TFP growth, the labor share and GDP in dollars is sourced from the Penn World Tables.

A particularly important variable is the interest-growth differential. It is constructed following ? as the difference between the effective interest rate and the growth rate of nominal GDP. The effective interest rate is composed of two parts: the actual interest spending divided by the average of the value of the stock of public debt in the current year and the previous year, and a depreciation adjustment that takes into account the part of public debt that is denominated in foreign currency³.

Since the interest here lies on the long-run low-frequency effects of demographics on the differential and the fiscal space, special attention was given to the handling of extreme values of the interest-growth differential. Table 1 shows all quartiles of its distribution:

Note that the distribution for the interest-growth differential spans an extremely wide range. It even gets close to -100% in the case of emerging economies, which means that the debt-to-GDP ratio gets wiped out the following year if the country maintains a zero primary

¹Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Israel, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom and United States.

²Argentina, Bolivia, Brazil, Bulgaria, Chile, China, Colombia, Costa Rica, Korea, Dominican Republic, Honduras, Hungary, India, Indonesia, Mexico, Nicaragua, Panama, Paraguay, Peru, Philippines, Poland, Romania, Thailand.

³Mathematically the effective interest rate is given by $\tilde{r}_t = r_t + \omega_{t-1}d_t$, where ω_{t-1} represents the last period's share of public debt denominated in foreign currency in the total and d_t is the country's domestic currency depreciation against the U.S. dollar.

	Advanced	Emerging
Minimum	-76.944	-99.985
1st quartile	-5.055	-15.207
Median	-1.023	-7.050
Mean	-1.960	-10.997
3rd quartile	1.593	-2.544
Maximum	19.925	50.570
Standard Deviation	6.621	15.279

Table 1: Empirical distribution: interest-growth differential (%)

balance. Such extreme values are the possible result of many causes, among them one can mention high inflation and financial repression on the negative side and high risk, strong currency depreciation or devaluation and output contraction on the positive side⁴. Yet, the present analysis hopes to get a sense of the fiscal effects these economies are experiencing and will experience due to their own demographic transition and the capital flows influenced by the demographic transitions abroad.

Demographic processes are slow-moving phenomena that despite structurally affecting the economy, clearly do not have any bearing in determining such extreme values for the interest-growth differential. Given that such values will affect the estimates, the task is then to find a range of values that can be considered “normal”, i.e. not affected by the problems mentioned above.

The proposed solution here is to consider the distribution of the interest-growth differential of advanced economies as the benchmark of a “normal” economy, and only consider the observations where the interest-growth differential falls between two standard-deviations of the mean of the distribution of *advanced* economies. The result is a range that starts at -16% and ends at 11% .

3.2 Stationarity Tests

In order to carry out the empirical analysis, it is necessary to evaluate the nature of each variable at hand with respect to stationarity. Table 4 shows the p-value of the ADF tests for each one of the variables and for all countries in the database. The ADF tests were carried out including a constant and choosing the number of included lags according to the Akaike criterion.

The tests show us that when treated independently, the interest-growth differential, the

⁴Mauro and Zhou (2021) exclude extreme events such as years with external sovereign default, inflation greater than 100% a year, and the top 1 percentile of exchange rate depreciation of the whole sample.

primary balance that stabilizes the debt-to-GDP ratio, and the TFP growth rate are the variables for which we reject the null hypothesis of the ADF test more often than not with 28, 29, and 43 of such cases out of a total of 46, respectively. The Im, Pesaran and Shin's W-statistic also point out to these variables as being stationary.

On the other hand, the growth rate of the working-age population, life expectancy at 65, the debt-to-GDP ratio and the labor share are clearly non-stationary with only 4, 0, 8, and 4 rejections of the null, respectively, and also a non-rejection by the W-statistic. This result suggests the presence of a unit root in each one of these series.

Carrying out the ADF test assuming independence across countries allows us to capture unit roots in variables of each cross-section unit, which can even be cointegrated, but it makes us miss the cointegration *across* cross-section units that can potentially be found in a panel data context. Therefore, it is useful to look for common factors across units and to assess their stationarity.

That is the goal of [Bai and Ng \(2004\)](#)'s PANIC stationarity test. In short, it decomposes a variable into a common and an idiosyncratic component using a factor model. The common factor captures the share trend across all panel units, while the idiosyncratic component represent the individual behavior that is specific to each panel unit. It then tests each component for stationarity using an ADF test.

Table 5 shows the results of three different PANIC tests. The first column considers all panel units in the same test, the second column considers only advanced economies and the third column considers only emerging economies. The numbers shown for each country are the p-values of the ADF test of their idiosyncratic factor.

and finds three common factors, all of them nonstationary

When there is cross-sectional dependence, it is useful to decompose the variable into two components: a vector of common factors to all cross-section units, F_{zt} , and a idiosyncratic element, ε_{zit} . In such case, a variable can present a unit root stemming either from the common factors or from its idiosyncratic component:

$$z_{it} = d_{it} + \beta'_{zi} F_{zt} + \varepsilon_{zit} \quad (28)$$

where

$$\begin{aligned} (1 - L)F_{zt} &= C(L)u_{zt} \\ \varepsilon_{zit} &= \rho\varepsilon_{zit-1} + \nu_{zit} \end{aligned}$$

In order to perform the decomposition and test both components for nonstationarity, we can resort to [Bai and Ng \(2004\)](#)'s PANIC test. The test follows a series of steps to decompose

each panel variable and to test for stationarity.

3.3 Methods

Given that the series are nonstationary, regular OLS estimation provides superconsistent estimates. A crucial shortcoming with performing an OLS estimation, however, is that the estimators are not normally distributed asymptotically and thus associated standard errors are not suitable for statistical inference on the cointegrating vector. They may yield spurious results. A straightforward approach that correct for this issue is the fully-modified OLS estimator (FMOLS) formulated by [Phillips and Hansen \(1990\)](#) and applied to panel data settings by [Pedroni \(1996\)](#) and [Phillips and Moon \(1999\)](#).

Fully-modified OLS uses a semi-parametric procedure to adjust the cointegrating variables and address the long-run correlation between the innovations in the regressors and the errors of the cointegrating equation. By doing so, it corrects the standard errors and allows for the use of standard t and F-statistics for hypothesis testing. It first implements a preliminary estimation of the cointegrating relationship using OLS for each cross-sectional unit i :

$$z_{it} = \hat{\alpha}_i + \hat{\beta}_{OLS,i} X_{it} + \hat{\epsilon}_{it} \quad (29)$$

The residuals $\hat{\epsilon}_{it}$ allow for an estimate of the long-run covariance matrices, Σ , Λ , and Ω using the Newey-West estimator. These estimates are then used to modify the data. The FMOLS estimator is then computed as:

$$\hat{\beta}_{FMOLS} = \left(\sum_i \sum_t \bar{X}_{it} \bar{X}_{it}' \right)^{-1} \left(\sum_i \sum_t \bar{X}_{it} \bar{z}_{it}^+ - T \hat{\lambda}_{12}^+ \right) \quad (30)$$

where the bar on top of the variables indicates that it has been time-demeaned and:

$$\bar{z}_{it}^+ = \bar{z}_{it} - \hat{\omega}_{12} \hat{\Omega}_{22}^{-1} \Delta X_{it} \quad (31)$$

$$\hat{\lambda}_{12}^+ = \hat{\lambda}_{12} - \hat{\omega}_{12} \hat{\Omega}_{22}^{-1} \hat{\omega}_{21} \quad (32)$$

The adjustments allow the asymptotic distribution of the FMOLS estimator to be centered around zero and the t-statistic to be asymptotically normal.

4 Results

4.1 Aging and the Interest-Growth Differential

Section 2 established the critical role of aging on affecting the interest-growth differential, one of the channels through which demographic shifts influence fiscal space. The interest-growth differential, defined as the difference between the interest rate on public debt and the growth rate of the economy, reacts differently depending on the source of aging: it contracts with increases in longevity but expands when fertility declines. These opposing effects emphasize the necessity of exploring this channel, particularly in the context of domestic aging.

In seeking to support the theoretical insights provided in the preceding chapter, this section employs an empirical analysis. The goal is to validate the observed relationships and ascertain their magnitude, extending the understanding to the data. The approach involves a series of regressions with the interest-growth differential as the dependent variable. These regressions are the estimation of the long-run relationship below.

$$d_{it} = \rho d_{it-1} + z'_{it}\beta + \bar{z}'_{-it}\gamma + X'_{it}\theta_0 + \bar{X}'_{-it}\theta_1 + u_{it} \quad (33)$$

where d_{it} represents the interest-growth differential and the autoregressive term d_{it-1} seeks to capture all sorts of persistent shocks the differential might be subject to and thus eliminate possible autocorrelation in the residuals, which can lead incorrect estimates of the variance of the estimators and thus incorrect inference.

Country i 's demographic variables are subsumed in vector $z_{it} = [g_{Nit}, \ell_{it}]$, where g_{Nit} is the growth rate of the working-age population, and ℓ_{it} represents life expectancy at 65. $\bar{z}_{-it}$ is the cross-section average of demographic variables of the other countries weighted by their GDP; X_{it} represents the domestic controls: labor share and TFP growth; and $\bar{X}_{-it}$ represents the cross-section average of foreign controls weighted by countries' GDP: labor share, TFP growth, and the debt-to-GDP ratio. FMOLS time-demeans all regressors.

The estimates are calculated using the FMOLS approach as explained in Section ?? above. Table 2 presents the results.

Equation (1) starts with a simple specification that includes both domestic demographic drivers to verify if the expected relationship holds even when controls are not included. Working-age population growth has a significant negative impact on the dependent variable, which aligns with the theoretical prediction. However, the positive and significant coefficient on longevity contradicts the theoretical expectation.

In regression (2), domestic controls are added but they do little to change the effects of demographic variables. The negative coefficient for working-age population growth becomes

Table 2: Interest-Growth Differential (1956-2018)

Dependent Variable: Differential Estimation Method	(1) P-FMOLS	(2) P-FMOLS	(3) P-FMOLS	(4) P-FMOLS	(5) Fixed Effects	(6) P-FMOLS	(7) P-FMOLS	(8) P-FMOLS
Differential (-1)	0.6593 (0.0000)***	0.6095 (0.0000)***	0.6427 (0.0000)***	0.5581 (0.0000)***	0.4392 (0.0000)***		0.6899 (0.0000)***	0.4228 (0.0000)***
Working-Age Population Growth (%)	-0.2631 (0.0531)*	-0.5259 (0.0000)***	-0.5017 (0.0004)***	-0.7624 (0.0000)***	-0.8792 (0.0000)***	-1.504 (0.0000)***	-0.5568 (0.0000)***	-1.1158 (0.0001)***
Longevity at 65 (years)	0.2798 (0.0000)***	0.2548 (0.0000)***	0.1446 (0.2104)	-0.0757 (0.4128)	0.0016 (0.9918)	-0.1153 (0.3253)	-0.1301 (0.2704)	-0.346 (0.039)**
Avg Advanced WA Pop. Growth (%)			1.4903 (0.0000)***	2.371 (0.0000)***	2.6259 (0.0000)***	4.7691 (0.0000)***	1.0657 (0.031)**	2.4979 (0.0094)***
Avg Advanced Longevity at 65 (years)			0.2761 (0.0182)**	-0.2995 (0.0433)**	-0.2662 (0.2478)	0.3713 (0.0489)**	-0.644 (0.0001)***	0.739 (0.011)**
Government Debt (%GDP)		-0.0021 (0.4602)		-0.0018 (0.4733)	-0.0004 (0.9396)	0.0031 (0.3246)	-0.0018 (0.4733)	-0.02 (0.0006)***
TFP Growth (%)		-0.501 (0.0000)***		-0.4806 (0.0000)***	-0.437 (0.0000)***	-0.4515 (0.0000)***	-0.5819 (0.0000)***	-0.4196 (0.0000)***
Labor Share (%)		0.0485 (0.0149)**		0.053 (0.0019)***	0.0546 (0.0937)*	0.0738 (0.0005)***	0.0085 (0.7103)	0.0428 (0.0947)*
Avg Advanced Govt Debt (%GDP)				0.0408 (0.0000)***	0.0494 (0.0000)***	0.088 (0.0000)***	0.0161 (0.1403)	0.0514 (0.0049)***
Avg Advanced TFP Growth (%)				-0.4421 (0.0000)***	-0.6043 (0.0000)***	0.2841 (0.0084)***	-0.6687 (0.0000)***	0.0027 (0.9871)
Avg Advanced Labor Share (%)				-0.5412 (0.0000)***	-0.4464 (0.0076)***	-0.3332 (0.04)**	-0.8679 (0.0000)***	0.3686 (0.1201)
Countries	All	All	All	All	All	All	Advanced	Emerging
Panel Units	46	46	46	46	46	46	23	23
Observations	2203	2168	2203	2168	2187	2197	1365	803
R-squared	0.4969	0.5848	0.5111	0.6138	0.6228	0.4342	0.6917	0.4572

p-values in parentheses, * *p* < 0.10, ** *p* < 0.05, *** *p* < 0.01

more pronounced and remains significant, but the positive sign on longevity remains. As predicted by the theory an increase in TFP growth shrinks the differential, while an increase in the labor share increases the interest rate and decreases GDP growth, leading to an increase in the differential.

Regression (3) introduces foreign demographic variables, the average advanced working-age population growth and average advanced longevity, but does not control for other relevant variables. It is interesting that both domestic and foreign working-age population growth come out significant and with the expected signs, while the signs of estimates for domestic and foreign longevity have the opposite sign to what is expected, with domestic longevity no longer being significant.

Regression (4) is the main specification and includes all relevant controls, both domestic and foreign. The signs of the effects of all demographic variables are consistent with the theory,

but the sign on longevity is insignificant⁵. The estimate of the coefficient on working-age population growth rate increases as controls are added, and in this specification, it implies that the effect of a decrease of 1 percentage point in the growth rate leads to an immediate increase in the differential of 0.76 percentage points and, considering the persistence of the shock, a long-term total increase of 1.73 percentage points ($0.7624/(1-0.5581)$). The (statistically insignificant) coefficient on longevity implies that one more year of life expectancy at 65 leads to an immediate fall in the differential of 0.076, and a long-term fall of 0.17. Given how many percentage points the working-age population growth rate falls across time vis-a-vis how many years life expectancy rises, the fall in working-age population growth dominates the increase in longevity. The overall effect of domestic population aging on the differential is to expand it, thus leading to a negative fiscal effect.

The coefficients on foreign demographics present a much larger effect on the differential. A decrease of 1 percentage point in the average working-age population growth rate in advanced economies leads to a short-term fall of 2.37 percentage points in the differential, and a long-term fall of 5.37 percentage points. An increase of one more year in the average longevity at 65 in advanced economies leads to an immediate reduction of 0.3 percentage points in the differential and a long-term fall of 0.68 percentage points. Here, both effects work in the same direction. Considering that the demographic transition in the average advanced economy saw from 1980 to 2018 a fall in working-age population growth rate of 1.1 percentage points and an increase in longevity at 65 of 4.4 years, the total long-term impact on the differential implied by the estimates is a reduction of 8.9 percentage points, a sizeable effect.

The complete specification is estimated again using panel fixed effects estimation in equation (5). This is a sanity check to see what one of the benchmark panel estimation procedures returns. The estimation clusters standard errors at the country level. Its estimates are not far off from the Pooled FMOLS estimation, which is expected, since the goal of P-FMOLS is to produce consistent estimates of the standard errors. The consistent standard errors are even smaller than the clustered standard errors of the panel fixed effects estimation, which are the ones that risk leading us to assume spurious relationships.

Regression (6) serves as a robustness check by excluding the lagged dependent variable. The effects of all variables should be compared to the total long-run effects implied by the estimates in equation (4). Domestic demographics show a similar effect as in (4), but slightly weaker. Estimates on foreign demographics again appear stronger than the domestic estimates but weaker than in (4). The sign on foreign longevity flips, however, and is at odds with the

⁵Filtering the data further by (arbitrarily) excluding years when a country saw a positive or negative change of more than 10 percentage points in its differential leads to significant and negative estimates for longevity, and the other results are maintained. Given the arbitrary aspect of this constraint, I report these regressions in the Appendix.

theory.

Equations (7) and (8) repeat the estimation in number (4), but only include advanced and emerging countries, respectively. In the case of advanced economies, the results are consistent with the full sample specification. However, the marginal effect of longevity increases vis-a-vis population growth. The domestic long-term effect of a fall of 1 percentage point in working-age population growth is 1.8 (versus 1.73 in (4)), while the long-run effect coming from an increase of one year in longevity is -0.42 (versus -0.17 in (4), albeit both statistically insignificant). The foreign demographic effects also show the same pattern, with foreign population growth losing its relative importance. The long-term effect of a fall in population growth in advanced economies is 3.44 (versus 5.37 in (4)), and the long-term effect of a rise in longevity abroad is 2.1 (versus 0.68 in (4)). Moreover, foreign demographics gain even more prominence here. Considering the average demographic changes on advanced countries from 1980 to 2018, the overall effect of foreign aging is a reduction of almost 13 percentage points in an advanced country's differential.

The estimated marginal effects of population growth in Table 2 are in line with the estimates found in the empirical investigation of [Checherita-Westphal and Semeano \(2020\)](#). In that study the authors investigate a panel of euro area countries and find that a fall of 1 percentage point in population growth leads to a more than proportional expansion in the differential. The authors, however, add the dependency ratio as another explanatory variable instead of longevity. The negative marginal effect they find for the dependency ratio ultimately captures the negative marginal effect of longevity on the differential, but the interpretation becomes much more difficult, as the authors duly note.

4.2 Aging and Fiscal Spillovers in Advanced and Emerging Economies

Finding a sufficient statistic that captures the essence of fiscal space implied by Definition 1 is not straightforward.

One correlated measure is the primary balance that stabilizes the current debt-to-GDP ratio assuming the interest-growth differential to be constant at its current level. The definition of the stabilizing primary balance, $\bar{\sigma}_t$ is:

$$\sigma_{it}^* \equiv b_{it} \left(\frac{R_{it}}{g_{Yit}} - 1 \right) \quad (34)$$

where b_{it} is the current debt-to-GDP ratio and R_t/g_{Yit} is the interest-growth differential.

The interest-growth differential of interest here is the one that is relevant to debt dynamics. As such, the interest rate is the implicit interest rate calculated as the ratio between interest spending and the debt stock. That provides the implicit one-year interest rate on government

debt. The output growth rate is the growth rate of nominal GDP measured in local currency.

The long-run relationship between the stabilizing primary balance and demographics will be estimated by the following equation:

$$\sigma_{it}^* = \alpha_i + d'_{it}\beta + \bar{d}'_{-it}\gamma + X'_{it}\theta_0 + \bar{X}'_{-it}\theta_1 + u_{it} \quad (35)$$

where $d_{it} = [g_{Nit}, \ell_{it}]$ is the vector of country i 's demographic variables: working-age population growth, g_{Nit} , and life expectancy at 65, ℓ_{it} ; $\bar{d}_{-it}$ is the cross-section average of demographic variables of the other countries weighted by their GDP; X_{it} represents the domestic controls: labor share and TFP growth; and $\bar{X}_{-it}$ represents the cross-section average of foreign controls weighted by countries' GDP: labor share, TFP growth, and the debt-to-GDP ratio. FMOLS time-demeans all regressors.

Table 3 shows the estimation of equation (35) for the full sample of emerging economies, from 1956 to 2019. Equations (1) through (6) estimate equation (35) using Panel FMOLS assuming that the marginal effects are homogeneous across countries. They come in two different flavors: with and without the autoregressive term. The addition of the AR(1) term seeks to capture the persistence of shocks to the dependent variable.

Equations (1) and (2) only include both demographic drivers as regressors. The growth rate of the working-age population is already statistically significant in this simple specification. The long-run effect is one-to-one, as in equation (1) an increase in the working-age population growth leads to a fall of $0.2588/(1 - 0.7496) = 1.03$ percentage points in the primary balance needed to stabilize the debt-to-GDP ratio. Equation (2) is similar, with a fall of 0.94 percentage points for the same change in the working-age population growth.

Equations (3) and (4) control for domestic TFP growth and for the domestic labor share, but do not differ much from the results in (1) and (2). The long-run effect of the working-age population growth is still nearly one-to-one. On the other hand, when foreign controls are added in equations (5) and (6) both domestic demographic drivers come out significant. The foreign controls utilized are the average of working-age population growth, life expectancy at 65, TFP growth, debt-to-GDP ratio and labor share across all advanced economies weighted by real GDP in 2017 dollars. Here the long-run effect of working-age population growth is lower: -0.77 in specification (5) and -0.51 in specification (6). However the effect of longevity is much more pronounced: one more year of life expectancy at 65 translates to a fall of 0.66 percentage points in the stabilizing primary balance in equation (5) and a fall of 0.24 percentage points in equation (6). This is particularly strong once one considers that over time life expectancy increased 4 years for every 1 percentage point fall in working-age population growth. In addition, the effect of longevity on fiscal space is ambiguous in theory,

Table 3: Debt-to-GDP Stabilizing Balance (1956-2018)

Dep. Variable: Stabilizing Balance	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Estimation Method	P-FMOLS	P-FMOLS	P-FMOLS	P-FMOLS	Fixed Effects	P-FMOLS	P-FMOLS	P-FMOLS
Stabilizing Balance (-1)	0.6496 (0.0000)***	0.6026 (0.0000)***	0.6402 (0.0000)***	0.5788 (0.0000)***	0.4314 (0.0000)***		0.7013 (0.0000)***	0.4754 (0.0000)***
Working-Age Population Growth (%)	-0.1573 (0.0768)*	-0.2548 (0.0006)***	-0.2353 (0.0082)***	-0.3299 (0.0000)***	-0.3953 (0.003)***	-0.6032 (0.0000)***	-0.2935 (0.0001)***	-0.2989 (0.046)**
Longevity at 65 (years)	0.0769 (0.005)***	0.0835 (0.0019)***	0.0898 (0.2205)	0.017 (0.7578)	0.0359 (0.7136)	0.018 (0.8323)	-0.109 (0.1659)	-0.0786 (0.3886)
Avg Advanced WA Pop. Growth (%)			0.5738 (0.0127)**	0.8415 (0.0025)***	1.2263 (0.0047)***	2.1301 (0.0000)***	0.3742 (0.2562)	0.0816 (0.8764)
Avg Advanced Longevity at 65 (years)			0.0365 (0.6208)	-0.2952 (0.0009)***	-0.2848 (0.0399)**	0.0474 (0.7292)	-0.3832 (0.0007)***	-0.1347 (0.372)
Labor Share (%)		0.0311 (0.0108)**		0.0316 (0.002)***	0.0373 (0.0729)*	0.0444 (0.0041)***	-0.0027 (0.8624)	0.0427 (0.0024)***
TFP Growth (%)		-0.1824 (0.0000)***		-0.1769 (0.0000)***	-0.1678 (0.0000)***	-0.1584 (0.0000)***	-0.2704 (0.0000)***	-0.1125 (0.0000)***
Avg Advanced Government Debt (%GDP)				0.0199 (0.0009)***	0.0321 (0.0005)***	0.0553 (0.0000)***	0.0076 (0.3006)	0.0247 (0.0133)**
Avg Advanced TFP Growth (%)				-0.3094 (0.0000)***	-0.3175 (0.0000)***	0.026 (0.7396)	-0.4093 (0.0000)***	-0.1496 (0.0935)*
Avg Advanced Labor Share (%)				-0.2409 (0.0014)***	-0.1865 (0.0389)**	0.08761 (0.4575)	-0.4420 (0.0000)***	0.0411 (0.7496)
Countries	All	All	All	All	All	All	Advanced	Emerging
Panel Units	46	46	46	46	46	46	23	23
Observations	2197	2168	2197	2168	2187	2197	1365	803
R-squared	0.4791	0.5398	0.4876	0.5626	0.5968	0.321	0.6603	0.4526

p-values in parentheses, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

which makes this result even surprising.

The point estimates of foreign demographics in equation (5) confirm the analytical results. A fall of 1 percentage point in the foreign working-age population growth lead to a long-term fall of 0.61 percentage points in the stabilizing balance, while a one-year increase in foreign life expectancy at 65 leads to a fall of 0.27 percentage points in the stabilizing primary balance.

Equations (7) and (8) carry out the same estimation as (5) and (6), but use a panel fixed effects estimation with standard errors clustered at the country level as a benchmark for comparison. The estimates are not very significant for most demographic variables even though their signs and magnitudes are similar to equations (5) and (6).

5 Conclusion

The demographic transition has been an ongoing slow-moving process for advanced economies that picked up especially in the post war era. Emerging economies have sustained a younger age structure of the population for a longer time and are now at the early stages of the aging process. The non-synchronized character of this transition has implications for emerging economies as it drives capital flows from the aging economies to the younger ones.

This study finds that emerging economies were affected positively by the aging of advanced economies starting the nineties, as financial integration increased between the developed and developing worlds.

It also finds that the different drivers of demographic change has opposing fiscal effects: a fall in the domestic working-age population growth is negative and calls for a fiscal adjustment, whereas an increase in domestic longevity increases savings and can thus lead to a positive fiscal effect.

Projections of future population aging by the United Nations ([United Nations, 2022](#)) show a very rapid fall in emerging countries' growth rate of working-age population, much faster than the one experienced by advanced economies. This suggests that as emerging economies age and catch up with advanced economies, they will be hit by a double-whammy of rising deficits and the abatement of capital flows which will critically tighten their fiscal space.

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Appendices

A Proofs of Propositions

A.1 Proposition 1

Proof.

Part 1: It suffices to show that the elasticity of the real interest rate with respect to the population growth rate is smaller than the elasticity of the output growth rate to the population growth rate:

$$\frac{\partial R_t}{\partial g_{Nt}} \frac{g_{Nt}}{R_t} < \frac{\partial g_{Yt}}{\partial g_{Nt}} \frac{g_{Nt}}{g_{Yt}} \quad (36)$$

Given the relationships:

$$\frac{\partial R_t}{\partial g_{Nt}} \frac{g_{Nt}}{R_t} = (\alpha - 1) \frac{\partial k_t}{\partial g_{Nt}} \frac{g_{Nt}}{k_t} \quad \text{and} \quad \frac{\partial g_{Yt}}{\partial g_{Nt}} \frac{g_{Nt}}{g_{Yt}} = 1 + \alpha \frac{\partial k_t}{\partial g_{Nt}} \frac{g_{Nt}}{k_t} \quad (37)$$

we can deduce:

$$\frac{\partial R_t}{\partial g_{Nt}} \frac{g_{Nt}}{R_t} < \frac{\partial g_{Yt}}{\partial g_{Nt}} \frac{g_{Nt}}{g_{Yt}} \iff \frac{\partial k_t}{\partial g_{Nt}} \frac{g_{Nt}}{k_t} > -1 \quad (38)$$

This implies that we only need to verify whether the capital stock in efficiency units is inelastic to the population growth rate. Applying the implicit function theorem to the market clearing equation (14) yields:

$$\frac{\partial k_t}{\partial g_{Nt}} \frac{g_{Nt}}{k_t} = - \frac{g_{At} g_{Nt}}{\frac{1-\alpha}{\alpha} \left(\frac{\delta_t}{1+\beta\delta_t} \nu g_{At} + \varphi_{t-1}^* \frac{\delta_t^*}{1+\beta^* \delta_t^*} \nu^* g_{At}^* \right) + (g_{At} g_{Nt} + \varphi_{t-1}^* g_{At}^* g_{Nt}^*)} \quad (39)$$

As long as $\varphi_{t-1}^* > 0$ and/or $\nu \neq 0$ (i.e. a sizeable foreign country exists and/or the pension replacement rate is not zero), the capital stock in efficiency units will be inelastic to the population growth rate and, as a result, the interest-growth differential will expand when the population growth rate falls.

Part 2: Here, the outcome is contingent on current savings. Similar to the previous part, it suffices to show that the elasticity of the real interest rate with respect to longevity is greater than the elasticity of the output growth rate to longevity:

$$\frac{\partial R_t}{\partial \delta_t} \frac{\delta_t}{R_t} < \frac{\partial g_{Yt}}{\partial \delta_t} \frac{\delta_t}{g_{Yt}} \quad (40)$$

Given that:

$$\frac{\partial R_t}{\partial \delta_t} \frac{\delta_t}{R_t} = (\alpha - 1) \frac{\partial k_t}{\partial \delta_t} \frac{\delta_t}{k_t} \quad \text{and} \quad \frac{\partial g_{Yt}}{\partial \delta_t} \frac{\delta_t}{g_{Yt}} = \alpha \frac{\partial k_t}{\partial \delta_t} \frac{\delta_t}{k_t} \quad (41)$$

We obtain as a result:

$$\frac{\partial R_t}{\partial \delta_t} \frac{\delta_t}{R_t} < \frac{\partial g_{Yt}}{\partial \delta_t} \frac{\delta_t}{g_{Yt}} \iff \frac{\partial k_t}{\partial \delta_t} \frac{\delta_t}{k_t} > 0 \quad (42)$$

The derivation for $\frac{\partial k_t}{\partial \delta_t}$ comes again from applying the implicit function theorem to the market clearing equation (14):

$$\begin{aligned} \frac{\partial k_t}{\partial \delta_t} &= \frac{\frac{\beta}{(1+\beta\delta_t)^2} ((1-\alpha) - \tau_{t-1}) k_{t-1}^\alpha - \frac{1-\alpha}{\alpha} \frac{1}{(1+\beta\delta_t)^2} \nu g_{At} k_t}{\frac{1-\alpha}{\alpha} \left(\frac{\delta_t}{1+\beta\delta_t} \nu g_{At} + \varphi_{t-1}^* \frac{\delta_t^*}{1+\beta^* \delta_t^*} \nu^* g_{At}^* \right) + (g_{At} g_{Nt} + \varphi_{t-1}^* g_{At}^* g_{Nt}^*)} \\ &= \frac{1}{\delta_t (1 + \beta \delta_t)} \frac{s_t / A_t}{\left[\frac{1-\alpha}{\alpha} \left(\frac{\delta_t}{1+\beta\delta_t} \nu g_{At} + \varphi_{t-1}^* \frac{\delta_t^*}{1+\beta^* \delta_t^*} \nu^* g_{At}^* \right) + (g_{At} g_{Nt} + \varphi_{t-1}^* g_{At}^* g_{Nt}^*) \right]} \end{aligned}$$

The inequality will thus depend on the sign of savings:

$$s_t \geq 0 \iff \frac{\partial k_t}{\partial \delta_t} \frac{\delta_t}{k_t} \geq 0 \iff \frac{\partial \frac{R_t}{g_{Yt}}}{\partial \delta_t} \leq 0 \quad (43)$$

□

As long as the replacement rate ν is low enough such that agents do not borrow against their future retirement benefits, we will have:

$$\frac{\partial \frac{R_t}{g_{Yt}}}{\partial \delta_t} < 0 \quad (44)$$

A.2 Proposition 2

Proof.

This proof follows the steps of the previous one. For both parts, we can compare the elasticities of the interest rate and output growth to foreign demographic variables.

Part 1: It is sufficient to demonstrate that:

$$\frac{\partial R_t}{\partial g_{Nt}^*} \frac{g_{Nt}^*}{R_t} > \frac{\partial g_{Yt}}{\partial g_{Nt}^*} \frac{g_{Nt}^*}{g_{Yt}} \quad (45)$$

We have:

$$\frac{\partial R_t}{\partial g_{Nt}^*} \frac{g_{Nt}^*}{R_t} = (\alpha - 1) \frac{\partial k_t}{\partial g_{Nt}^*} \frac{g_{Nt}^*}{k_t} \quad \text{and} \quad \frac{\partial g_{Yt}}{\partial g_{Nt}^*} \frac{g_{Nt}^*}{g_{Yt}} = \alpha \frac{\partial k_t}{\partial g_{Nt}^*} \frac{g_{Nt}^*}{k_t} \quad (46)$$

Thus, the inequality implies that it is sufficient to show that:

$$\frac{\partial k_t}{\partial g_{Nt}^*} \frac{g_{Nt}^*}{k_t} < 0 \quad (47)$$

The implicit differentiation of (14) with respect to g_{Nt}^* yields:

$$\frac{\partial k_t}{\partial g_{Nt}^*} \frac{g_{Nt}^*}{k_t} = - \frac{\varphi_{t-1}^* g_{At}^* g_{Nt}^*}{\frac{1-\alpha}{\alpha} \left(\frac{\delta_t}{1+\beta\delta_t} \nu g_{At} + \varphi_{t-1}^* \frac{\delta_t^*}{1+\beta^* \delta_t^*} \nu^* g_{At}^* \right) + (g_{At} g_{Nt} + \varphi_{t-1}^* g_{At}^* g_{Nt}^*)} < 0 \quad (48)$$

which always holds.

Part 2: It is enough to show that:

$$\frac{\partial R_t}{\partial \delta_t^*} \frac{\delta_t^*}{R_t} < \frac{\partial g_{Yt}}{\partial \delta_t^*} \frac{\delta_t^*}{g_{Yt}} \quad (49)$$

We have:

$$\frac{\partial R_t}{\partial \delta_t^*} \frac{\delta_t^*}{R_t} = (\alpha - 1) \frac{\partial k_t}{\partial \delta_t^*} \frac{\delta_t^*}{k_t} \quad \text{and} \quad \frac{\partial g_{Yt}}{\partial \delta_t^*} \frac{\delta_t^*}{g_{Yt}} = \alpha \frac{\partial k_t}{\partial \delta_t^*} \frac{\delta_t^*}{k_t} \quad (50)$$

Therefore, the inequality implies that:

$$\frac{\partial k_t}{\partial \delta_t^*} \frac{\delta_t^*}{k_t} > 0 \quad (51)$$

The elasticity above is symmetric to the elasticity with respect to domestic longevity, and therefore it holds as long as $s_t^* > 0$. $\square$

A.3 Proposition 3

Proof. The goal here is understanding how a change in an exogenous demographic variable, denoted as x , influences the interest-growth differential in future time periods. Develop the formula of the differential to find:

$$\frac{d \frac{R_{t+j}}{g_{Yt+j}} \frac{x}{R_{t+j}}}{dx} = \alpha \frac{dk_{t+j-1}}{dx} \frac{x}{k_{t+j-1}} - \frac{dk_{t+j}}{dx} \frac{x}{k_{t+j}}, \text{ for } j = 1, 2, \dots \quad (52)$$

Thus the key to determining the sign of this marginal effect lies in how x influences the capital stock. Using (14), let's define:

$$\begin{aligned} F_{t+j} \equiv & \left(\frac{\beta \delta_{t+j}}{1 + \beta \delta_{t+j}} ((1 - \alpha) - \tau_{t+j-1}) + \varphi_{t+j-1}^* \frac{\beta^* \delta_{t+j}^*}{1 + \beta^* \delta_{t+j}^*} ((1 - \alpha) - \tau_{t+j-1}^*) \right) k_{t+j-1}^\alpha \\ & - \frac{1 - \alpha}{\alpha} \left(\frac{\delta_{t+j}}{1 + \beta \delta_{t+j}} \nu + \varphi_{t+j-1}^* \frac{\delta_{t+j}^*}{1 + \beta^* \delta_{t+j}^*} \nu^* \right) k_{t+j} \\ & - (g_{At+j} g_{Nt+j} + g_{At+j}^* g_{Nt+j}^* \varphi_{t+j-1}^*) k_{t+j} - (b_{t+j} + \varphi_{t+j-1}^* b_{t+j}^*) j_{t+j-1}^\alpha = 0 \end{aligned} \quad (53)$$

And note that τ_{t+j-1} and τ_{t+j-1}^* are determined endogenously:

$$\tau_{t+j-1} = \gamma_{t+j-1} + \frac{\delta_{t+j-1} \nu (1 - \alpha)}{g_{Nt+j-1}} + \frac{\alpha k_{t+j-2}^\alpha}{g_{At+j-1} g_{Nt+j-1} k_{t+j-1}} b_{t+j-1} - b_{t+j} \quad (54)$$

and similarly for τ_{t+j-1}^* .

The implicit function theorem implies the relationship below:

$$\frac{dk_{t+j}}{dx} = -\frac{\frac{\partial F_{t+j}}{\partial k_{t+j-1}}}{\frac{\partial F_{t+j}}{\partial k_{t+j}}} \frac{dk_{t+j-1}}{dx} - \frac{\frac{\partial F_{t+j}}{\partial k_{t+j-2}}}{\frac{\partial F_{t+j}}{\partial k_{t+j}}} \frac{dk_{t+j-2}}{dx} - \frac{\frac{\partial F_{t+j}}{\partial x}}{\frac{\partial F_{t+j}}{\partial k_{t+j}}} \quad (55)$$

Here, x affects the capital stock only in the current and future periods. This lets us determine the immediate effect first (as it was done with Propositions 1 and 2) and then find the effect one period ahead. We can then use those to find the effect two periods ahead. Also note that the last term vanishes for the marginal effect at two or more periods ahead.

It can be shown that equation (55) implies:

$$\frac{dk_{t+j}}{dx} x = \alpha \frac{dk_{t+j-1}}{dx} \frac{x}{k_{t+j-1}} k_{t+j} + \frac{Z_{t+j}}{D_{t+j}} \frac{dk_{t+j-1}}{dx} \frac{x}{k_{t+j-1}} - \alpha^2 \frac{Z_{t+j}}{D_{t+j}} \frac{dk_{t+j-2}}{dx} \frac{x}{k_{t+j-2}} + \frac{\frac{\partial F_{t+j}}{\partial x}}{D_{t+j}} x \quad (56)$$

where:

$$Z_{t+j} = \left(\frac{\beta \delta_{t+j}}{1 + \beta \delta_{t+j}} \frac{\alpha k_{t+j-2}^\alpha}{g_{At+j-1} g_{Nt+j-1} k_{t+j-1}} b_{t+j-1} + \varphi_{t+j-1}^* \frac{\beta^* \delta_{t+j}^*}{1 + \beta^* \delta_{t+j}^*} \frac{\alpha k_{t+j-2}^\alpha}{g_{At+j-1}^* g_{Nt+j-1}^* k_{t+j-1}^*} b_{t+j-1}^* \right) k_{t+j-1}^\alpha > 0$$

$$D_{t+j} = \frac{1 - \alpha}{\alpha} \left(\frac{\delta_{t+j}}{1 + \beta \delta_{t+j}} \nu g_{At+j} + \varphi_{t+j-1}^* \frac{\delta_{t+j}^*}{1 + \beta^* \delta_{t+j}^*} \nu^* g_{At+j}^* \right) + (g_{At+j} g_{Nt+j} + \varphi_{t+j-1}^* g_{At+j}^* g_{Nt+j}^*) > 0$$

Note that subtracting the first term on the right-hand side from the left hand side gives us the negative of the elasticity of the differential with respect to x . This means that it is enough to check the sign of:

$$\frac{d \frac{R_{t+j}}{g_{Yt+j}}}{dx} \frac{x}{\frac{R_{t+j}}{g_{Yt+j}}} \geq 0 \iff \frac{Z_{t+j}}{D_{t+j}} \left(\frac{dk_{t+j-1}}{dx} \frac{x}{k_{t+j-1}} - \alpha^2 \frac{dk_{t+j-2}}{dx} \frac{x}{k_{t+j-2}} \right) + \frac{\frac{\partial F_{t+j}}{\partial x}}{D_{t+j}} x \leq 0 \quad (57)$$

Also note that since the last term vanishes for $j = 2, 3, \dots$, if the elasticity of the differential at $j = 1$ is negative, then all the future elasticities will also be negative, since the term in parentheses will be positive. Thus, it suffices to check that the sign of the marginal effect on the differential is negative for $j = 1$ to show that the entire impulse response function is negative. It is also enough to compute the derivatives with respect to just g_{Nt} and δ_t given the symmetry of the problem.

Part 1: Start with g_{Nt} . In this case we have:

$$\frac{\partial F_{t+1}}{\partial g_{Nt}} = \frac{\beta \delta_{t+1}}{1 + \beta \delta_{t+1}} \left(\frac{\delta_t \nu (1 - \alpha)}{g_{Nt}} + \frac{\alpha k_{t-1}^\alpha}{g_{At} g_{Nt} k_t} b_t \right) \frac{1}{g_{Nt}} k_t^\alpha \quad \text{and} \quad \frac{dk_{t-1}}{dg_{Nt}} \frac{g_{Nt}}{k_{t-1}} = 0 \quad (58)$$

which means that we just need to check the sign of the expression below:

$$Z_{t+1} \frac{dk_t}{dg_{Nt}} \frac{g_{Nt}}{k_t} + \frac{\beta\delta_{t+1}}{1 + \beta\delta_{t+1}} \left(\frac{\delta_t \nu(1 - \alpha)}{g_{Nt}} + \frac{\alpha k_{t-1}^\alpha}{g_{At} g_{Nt} k_t} b_t \right) k_t^\alpha \leq 0 \quad (59)$$

The expression above shows clearly the two opposing effects of changes in population growth on the differential one period ahead. The first term indicates the fiscal burden that comes from the higher interest rate: as population growth rises, the cost of debt rollover increases, an adverse fiscal impact. The second term captures the beneficial impact a higher population growth has on the primary balance as it expands the tax base and also on the debt rollover as it boosts growth. Expanding Z_{t+1} and rearranging leads to:

$$\left[\left(1 + \frac{dk_t}{dg_{Nt}} \frac{g_{Nt}}{k_t} \right) \frac{\beta\delta_{t+1}}{1 + \beta\delta_{t+1}} \frac{\alpha k_{t-1}^\alpha}{g_{At} g_{Nt} k_t} b_t + \frac{\beta\delta_{t+1}}{1 + \beta\delta_{t+1}} \frac{\delta_t \nu(1 - \alpha)}{g_{Nt}} + \left(\varphi_t^* \frac{\beta^* \delta_{t+1}^*}{1 + \beta^* \delta_{t+1}^*} \frac{\alpha k_{t-1}^\alpha}{g_{At}^* g_{Nt}^* k_t^*} b_t^* \right) \frac{dk_t}{dg_{Nt}} \frac{g_{Nt}}{k_t} \right] k_t^\alpha \leq 0 \quad (60)$$

The first and second terms are positive since Proposition 1 shows that the elasticity is between -1 and 0. The third term is negative. If b_t is big enough or either b_t^* or φ_t^* are small enough then the expression certainly positive and, as a result, the marginal effect of an increase in population growth on the differential next period is negative and so are all future marginal effects.

Part 2: Doing the same for δ_t , we have:

$$\frac{\partial F_{t+1}}{\partial \delta_t} = -\frac{\beta\delta_{t+1}}{1 + \beta\delta_{t+1}} \frac{\nu(1 - \alpha)}{g_{Nt}} k_t^\alpha \quad \text{and} \quad \frac{dk_{t-1}}{d\delta_t} \frac{\delta_t}{k_{t-1}} = 0 \quad (61)$$

The relevant expression then becomes:

$$Z_{t+1} \frac{dk_t}{d\delta_t} \frac{\delta_t}{k_t} - \frac{\beta\delta_{t+1}}{1 + \beta\delta_{t+1}} \frac{\delta_t \nu(1 - \alpha)}{g_{Nt}} k_t^\alpha \leq 0 \quad (62)$$

The expression above shows clearly the two opposing effects of changes in longevity on the differential one period ahead. The first term captures the fiscal impact that comes from the capital stock: as agents expect to live longer as save more, the cost of debt rollover falls, a beneficial fiscal impact. The second term captures the adverse fiscal impact higher longevity has on the primary balance as it increases the number of retirees receiving benefits.

$$\left[\left(\frac{\beta \delta_{t+1}}{1 + \beta \delta_{t+1}} \left(\frac{dk_t}{d\delta_t} \frac{\delta_t}{k_t} \frac{\alpha k_{t-1}^\alpha}{g_{At} g_{Nt} k_t} b_t - \frac{\delta_t \nu (1 - \alpha)}{g_{Nt}} \right) + \varphi_t^* \frac{\beta^* \delta_{t+1}^*}{1 + \beta^* \delta_{t+1}^*} \frac{dk_t}{d\delta_t} \frac{\delta_t}{k_t} \frac{\alpha k_{t-1}^\alpha}{g_{At}^* g_{Nt}^* k_t} b_t^* \right) \right] k_t^\alpha \leq 0 \quad (63)$$

In this context it is sufficient to ensure that the elasticity of the capital stock (in efficiency units) to longevity is high enough. Given that we are comparing the combination of the debt-to-GDP ratios in two countries to pension expenditures (as % of GDP) in only one of them, the condition should not be very restrictive. Just like in part 1, the higher the domestic stock of debt, the more elastic the capital stock is to longevity. Unlike part 1, though, a higher stock of foreign debt or a larger foreign country ensure that the inequality is positive. $\square$

A.4 Proposition 4

Proof.

Part 1: The effect of changes in population growth on the fiscal space is straightforward given that the impacts on the primary surplus and on the interest-growth differential go in the same direction.

The impact of a *fall* in the population growth rate on the primary balance is negative.

$$\sum_{i=0}^{\infty} \left(\prod_{j=0}^i \frac{g_{Y,t+j}}{R_{t+j}} \right) \frac{\partial \sigma_{t+i}^d}{\partial g_{Nt}} = \frac{g_{Y,t}}{R_t} \frac{\nu(1-\alpha)\delta_t}{g_{Nt}^2} > 0 \quad (64)$$

The impact on the interest-growth differential can be decomposed as follows:

$$\begin{aligned} \sum_{i=0}^{\infty} \frac{\partial \left(\prod_{j=0}^i \frac{g_{Y,t+j}}{R_{t+j}} \right)}{\partial g_{Nt}} \sigma_{t+i} &= \frac{\partial \frac{g_{Y,t}}{R_t}}{\partial g_{Nt}} \frac{g_{Nt}}{\frac{g_{Y,t}}{R_t}} \frac{g_{Y,t}}{R_t} \sigma_t + \\ &+ \left(\frac{\partial \frac{g_{Y,t}}{R_t}}{\partial g_{Nt}} \frac{g_{Nt}}{\frac{g_{Y,t}}{R_t}} + \frac{\partial \frac{g_{Y,t+1}}{R_{t+1}}}{\partial g_{Nt}} \frac{g_{Nt}}{\frac{g_{Y,t+1}}{R_{t+1}}} \right) \frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \sigma_{t+1} + \\ &+ \left(\frac{\partial \frac{g_{Y,t}}{R_t}}{\partial g_{Nt}} \frac{g_{Nt}}{\frac{g_{Y,t}}{R_t}} + \frac{\partial \frac{g_{Y,t+1}}{R_{t+1}}}{\partial g_{Nt}} \frac{g_{Nt}}{\frac{g_{Y,t+1}}{R_{t+1}}} + \frac{\partial \frac{g_{Y,t+2}}{R_{t+2}}}{\partial g_{Nt}} \frac{g_{Nt}}{\frac{g_{Y,t+2}}{R_{t+2}}} \right) \frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \frac{g_{Y,t+2}}{R_{t+2}} \sigma_{t+2} + \dots \end{aligned}$$

Now rearrange according to the changes in the interest-growth differential in each period:

$$\begin{aligned}
 \sum_{i=0}^{\infty} \frac{\partial \left(\prod_{j=0}^i \frac{g_{Y,t+j}}{R_{t+j}} \right)}{\partial g_{Nt}} \sigma_{t+i} &= \frac{\partial \frac{g_{Y,t}}{R_t}}{\partial g_{Nt}} \frac{g_{Nt}}{\frac{g_{Y,t}}{R_t}} \frac{1}{g_{Nt}} \left(\frac{g_{Y,t}}{R_t} \sigma_t + \frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \sigma_{t+1} + \frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \frac{g_{Y,t+2}}{R_{t+2}} \sigma_{t+2} + \dots \right) \\
 &+ \frac{\partial \frac{g_{Y,t+1}}{R_{t+1}}}{\partial g_{Nt}} \frac{g_{Nt}}{\frac{g_{Y,t+1}}{R_{t+1}}} \frac{1}{g_{Nt}} \left(\frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \sigma_{t+1} + \frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \frac{g_{Y,t+2}}{R_{t+2}} \sigma_{t+2} + \dots \right) \\
 &+ \frac{\partial \frac{g_{Y,t+2}}{R_{t+2}}}{\partial g_{Nt}} \frac{g_{Nt}}{\frac{g_{Y,t+2}}{R_{t+2}}} \frac{1}{g_{Nt}} \left(\frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \frac{g_{Y,t+2}}{R_{t+2}} \sigma_{t+2} + \frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \frac{g_{Y,t+2}}{R_{t+2}} \frac{g_{Y,t+3}}{R_{t+3}} \sigma_{t+3} + \dots \right) \\
 &+ \dots \\
 &= \frac{1}{g_{Nt}} \left(\frac{\partial \frac{g_{Y,t}}{R_t}}{\partial g_{Nt}} \frac{g_{Nt}}{\frac{g_{Y,t}}{R_t}} b_t + \frac{\partial \frac{g_{Y,t+1}}{R_{t+1}}}{\partial g_{Nt}} \frac{g_{Nt}}{\frac{g_{Y,t+1}}{R_{t+1}}} \frac{g_{Y,t}}{R_t} b_{t+1} + \frac{\partial \frac{g_{Y,t+2}}{R_{t+2}}}{\partial g_{Nt}} \frac{g_{Nt}}{\frac{g_{Y,t+2}}{R_{t+2}}} \frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} b_{t+2} + \dots \right) \\
 &> 0 \text{ using Propositions 1 and 2, if } b_{t+i} \geq 0, \forall i = 1, 2, \dots
 \end{aligned}$$

Part 2: The effect of changes in longevity on the fiscal space is ambiguous given that the impacts on the primary surplus and on the interest-growth differential go in different directions.

The impact of a rise in longevity on the primary balance is negative.

$$\sum_{i=0}^{\infty} \left(\prod_{j=0}^i \frac{g_{Y,t+j}}{R_{t+j}} \right) \frac{\partial \sigma_{t+i}^d}{\partial \delta_t} = - \frac{g_{Y,t}}{R_t} \frac{\nu(1-\alpha)}{g_{Nt}} < 0 \quad (65)$$

The impact on the interest-growth differential can be decomposed as in Part 1:

$$\begin{aligned}
 \sum_{i=0}^{\infty} \frac{\partial \left(\prod_{j=0}^i \frac{g_{Y,t+j}}{R_{t+j}} \right)}{\partial \delta_t} \sigma_{t+i} &= \frac{\partial \frac{g_{Y,t}}{R_t}}{\partial \delta_t} \frac{\delta_t}{\frac{g_{Y,t}}{R_t}} \frac{1}{\delta_t} \left(\frac{g_{Y,t}}{R_t} \sigma_t + \frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \sigma_{t+1} + \frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \frac{g_{Y,t+2}}{R_{t+2}} \sigma_{t+2} + \dots \right) \\
 &+ \frac{\partial \frac{g_{Y,t+1}}{R_{t+1}}}{\partial \delta_t} \frac{\delta_t}{\frac{g_{Y,t+1}}{R_{t+1}}} \frac{1}{\delta_t} \left(\frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \sigma_{t+1} + \frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \frac{g_{Y,t+2}}{R_{t+2}} \sigma_{t+2} + \dots \right) \\
 &+ \frac{\partial \frac{g_{Y,t+2}}{R_{t+2}}}{\partial \delta_t} \frac{\delta_t}{\frac{g_{Y,t+2}}{R_{t+2}}} \frac{1}{\delta_t} \left(\frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \frac{g_{Y,t+2}}{R_{t+2}} \sigma_{t+2} + \frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} \frac{g_{Y,t+2}}{R_{t+2}} \frac{g_{Y,t+3}}{R_{t+3}} \sigma_{t+3} + \dots \right) \\
 &+ \dots \\
 &= \frac{1}{\delta_t} \left(\frac{\partial \frac{g_{Y,t}}{R_t}}{\partial \delta_t} \frac{\delta_t}{\frac{g_{Y,t}}{R_t}} b_t + \frac{\partial \frac{g_{Y,t+1}}{R_{t+1}}}{\partial \delta_t} \frac{\delta_t}{\frac{g_{Y,t+1}}{R_{t+1}}} \frac{g_{Y,t}}{R_t} b_{t+1} + \frac{\partial \frac{g_{Y,t+2}}{R_{t+2}}}{\partial \delta_t} \frac{\delta_t}{\frac{g_{Y,t+2}}{R_{t+2}}} \frac{g_{Y,t}}{R_t} \frac{g_{Y,t+1}}{R_{t+1}} b_{t+2} + \dots \right) \\
 &> 0 \text{ using Propositions 1 and 2, if } b_{t+i} \geq 0, \forall i = 1, 2, \dots
 \end{aligned}$$

□

B Unit Root Tests

Table 4: Unit Root Tests - Independent Cross-Sections

	Differential	Stabilizing Balance	Population Growth	Life Expectancy	Debt-to-GDP	Labor Share	TFP Growth
Argentina	0.1969	0.4522	0.5666	0.5208	0.9382	0.9700	0.0585
Australia	0.0390	0.1453	0.1087	0.9998	0.2094	0.9230	0.0000
Austria	0.0132	0.0012	0.1976	0.9980	0.8296	0.6827	0.0000
Belgium	0.1869	0.3311	0.1683	0.9984	0.4458	0.7859	0.2398
Bolivia	0.0233	0.0120	0.0757	0.9988	0.3134	0.4853	0.0207
Brazil	0.0483	0.1609	0.9995	0.9743	0.3870	0.9680	0.0658
Bulgaria	0.3397	0.0893	0.9926	0.7257	0.0285	0.3833	0.0000
Canada	0.2829	0.3374	0.8436	0.9984	0.6270	0.8113	0.0000
Chile	0.0027	0.0000	0.0861	0.5406	0.0001	0.2648	0.0679
China	0.0076	0.1396	0.7908	0.9291	0.9999	0.4956	0.0163
Colombia	0.0280	0.0027	0.9643	0.9996	0.7207	0.2450	0.0073
Costa Rica	0.0028	0.0000	0.9995	0.8327	0.9072	-	0.0039
Korea	0.0082	0.0003	0.6474	0.9996	0.9849	0.1709	0.0000
Denmark	0.7697	0.5113	0.5659	0.9998	0.4174	0.6249	0.0000
Dominican Republic	0.0502	0.0106	0.8799	0.7963	0.6691	0.9207	0.0054
Finland	0.0089	0.0324	0.8418	0.9986	0.8678	0.8264	0.0000
France	0.5096	0.3555	0.5064	0.9866	0.9967	0.6620	0.0027
Germany	0.0216	0.1956	0.3852	0.9984	0.8387	0.4528	0.0010
Greece	0.1602	0.9950	0.6503	0.8625	0.9966	0.8945	0.0000
Honduras	0.0168	0.0060	0.5278	0.9997	0.8953	0.5658	0.0001
Hungary	0.2622	0.1583	0.9802	0.8393	0.0438	0.7251	0.0014
Iceland	0.1323	0.0040	0.4711	0.9985	0.4205	0.0228	0.0002
India	0.0001	0.0230	1.0000	0.8550	0.5489	0.9077	0.0000
Indonesia	0.0685	0.0327	0.7977	0.9886	0.0000	0.9694	0.0143
Ireland	0.0047	0.0101	0.7614	0.9990	0.0648	0.9996	0.0000
Israel	0.3320	0.3184	0.0175	0.7304	0.0411	0.7131	0.0000
Italy	0.1964	0.1826	0.1433	0.9920	0.8798	0.8367	0.3383
Japan	0.5218	0.7234	0.9043	0.9609	0.9947	0.8471	0.0002
Mexico	0.0402	0.0406	0.9794	0.5953	0.4405	0.6842	0.0000
Netherlands	0.1891	0.0736	0.9488	0.9949	0.2589	0.9117	0.0327
New Zealand	0.1077	0.0200	0.8673	1.0000	0.4928	0.8167	0.0000
Nicaragua	0.1757	0.3621	0.9996	0.5961	0.7942	0.8374	0.0145
Norway	0.0204	0.0939	0.3131	0.9999	0.0270	0.6220	0.0041
Panama	0.0006	0.0001	0.6047	0.9939	0.6747	0.6572	0.0003
Paraguay	0.1808	0.2194	1.0000	0.9916	0.1635	0.5343	0.0018
Peru	0.0469	0.0254	0.6955	0.9997	0.4627	0.9998	0.0231
Philippines	0.0432	0.0001	0.7099	0.9810	0.3053	0.0006	0.0000
Poland	0.0004	0.0001	0.9997	0.4503	0.6178	0.6392	0.0192
Portugal	0.0597	0.0253	0.5909	0.9969	0.9042	0.0000	0.0005
Romania	0.0082	0.0088	0.9999	0.4497	0.5440	0.1877	0.1035
Spain	0.5357	0.1978	0.2574	0.9789	0.5948	0.9562	0.0159
Sweden	0.0885	0.0368	0.1079	0.9998	0.3483	0.3748	0.0001
Switzerland	0.0598	0.0306	0.0511	0.9873	0.2795	0.0018	0.0000
Thailand	0.0082	0.0074	0.9194	0.9506	0.0873	1.0000	0.0000
United Kingdom	0.1963	0.0473	0.2044	0.9992	0.1943	0.8233	0.0000
United States	0.0013	0.0001	0.7999	0.9843	0.8451	0.6445	0.0000
IPS W-stat	0.0000	0.0000	1.0000	1.0000	0.5936	0.9982	0.0000

Table 5: Unit Root Tests - Dependent Cross-Sections

Idiosyncratic factors			
Countries	MCF	MCF Advanced	MCF Emerging
Argentina	0.0153	-	0.1357
Australia	0.0167	0.3321	-
Austria	0.0175	0.0221	-
Belgium	0.267	0.8477	-
Bolivia	0.4891	-	0.6003
Brazil	0.2324	-	0.5119
Bulgaria	0.9808	-	0.3904
Canada	0.0098	0.0239	-
Chile	0.9631	-	0.9272
China	0.526	-	0.4909
Colombia	0.5237	-	0.1604
Costa Rica	0.9999	-	0.3764
Korea	0.8805	-	0.0032
Denmark	0.7159	0.6539	-
Dominican Republic	0.9215	-	0.4308
Finland	0.0199	0.0002	-
France	0.0178	0	-
Germany	0.3259	0.0002	-
Greece	0.2945	0.1631	-
Honduras	0.7761	-	0
Hungary	0.2644	-	0.3804
Iceland	0.0004	0.141	-
India	0.0018	-	0.7319
Indonesia	0.9650	-	0.8442
Ireland	0.0405	0.0325	-
Israel	0.0000	0.0688	-
Italy	0.0365	0.002	-
Japan	0.9999	0	-
Mexico	0.4014	-	0.6527
Netherlands	0.0103	0.5154	-
New Zealand	0.9999	0	-
Nicaragua	0.9879	-	0.7819
Norway	0.1115	0.3983	-
Panama	0.0035	-	0.3492
Paraguay	0.5425	-	0.9909
Peru	0.0000	-	0.9996
Philippines	0.5637	-	0.1931
Poland	0.7385	-	0.0332
Portugal	0.9999	0.0593	-
Romania	0.0002	-	0.0017
Spain	0.0274	0.1182	-
Sweden	0.9185	0.6688	-
Switzerland	0.0011	0.0038	-
Thailand	0.1035	-	0.0038
United Kingdom	0.0394	0.0013	-
United States	0.0088	0.0123	-

Common factors			
Total number	3	1	8
Nonstationary	3	1	8
Pooled Statistic	0.0000	0.0000	0.0000
Cross-sections	46	23	23
Balanced Observations	19	32	21